Fiscal Year:	FY 2017	**Task Last Updated:**	FY 09/21/2017
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Project Title:	Burst Wave Lithotripsy Generator and Probe Development for Commercial Kidney Stone Comminution System		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	04/01/2016	**End Date:**	05/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMST00015		
Performance Goal No.:			
Performance Goal Text:			

Astronauts are at increased risk of nephrolithiasis (kidney stones) due to microgravity, dehydration, and altered bone metabolism associated with space flight, as well as medications prescribed to mitigate visual impairment and intracranial pressure (VIIP). A stone can cause debilitating pain as it passes, or worse, become obstructing, leading to the progression of urinary tract infection, sepsis, renal failure, and death. Kidney stone disease is a recognized risk for astronauts and is listed in the ExMC Gap Report 4.13 Limited capability to diagnose and treat a renal stone. Break Wave Lithotripsy (BWL or Break Wave) will fragment large kidney stones more quickly, safely, cheaply, and effectively compared to existing FDA approved methods and with the goal of doing so without the need for general anesthesia or ionizing X-ray radiation. Eliminating the need for fluoroscopy, general anesthesia, and a surgical facility will have a dramatic impact on the delivery of lithotripsy. Furthermore, the system is projected to be comparable in cost and size to an ultrasound imaging system, making it feasible for use in clinics, emergency rooms, and in outer space. The overall

Task Description:

goal of this grant was to design, build, and test a therapeutic ultrasound Break Wave system with specific emphasis on creating a system that is compact, lightweight, and low power. Specifically, the aims of the project were to: • Aim 1: Fabricate an electronic generator and probe for Break Wave treatment with specific focus on a Space oriented system that is compact, lightweight, and power-efficient. • Aim 2: Demonstrate the ability to break kidney stones using the compact generator and probe developed in Aim 1. • Aim 3: Extend generator capabilities to drive both Break Wave and stone pushing based on findings and experience from Aims 1 and 2.
Key Findings: The project successfully achieved its goals. Below is a summary of our key achievements:

• Successfully redesigned the Break Wave Generator to a size and mass that is similar to computer DVD drive. The Generator volume and mass was reduced by a factor of 10x from its original form.

• Reduced the power requirement of the Generator such that it can be driven by a standard 100W medical grade power supply. This power supply is commercially available and it is the size of a pack of cards.

• Extended the Generator pulse generation capabilities to both reposition and break stones.

• Simulated, designed, and fabricated a four channel Break Wave therapeutic probe that is only 85 cm in diameter and which can accommodate a standard phased array imaging probe for procedure guidance.

• Demonstrated the ability to break 3 mm to 10 mm human kidney stone in-vitro.

• Redesigned the Break Wave architecture to allow for integration with the NASA flexible ultrasound system (FUS) or any commercially available diagnostic ultrasound system. Meaning, we should be able to integrate with any diagnostic ultrasound system that NASA chooses to fly on long duration space mission.

Impact: The goal of the project was to create a Break Wave system (generator and probe) that addresses space based needs such as a system that is compact, lightweight, and power-efficient. While these characteristics are desirable to facilitate integration of the Break Wave components into a commercial Earth-bound ultrasound imaging system chassis, they are critically important for NASA and space bound missions. Per our goals, we successfully designed, built, and tested a miniaturized BWL system that is compact, lightweight, and power-efficient. The system was also re-architected to ease integration with the NASA FUS or any other diagnostic ultrasound system that NASA chooses to fly in space. With the achievements of this project, we are closer to significantly reducing the risk associated with kidney stones on long term space missions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Kidney and ureteral stones are one of the most painful medical conditions. The lifetime incidence of kidney stones is nearly 13 percent in men and 7 percent in women. The likelihood of recurrence is 50 percent or greater at five years and up to 80 percent at 10 years. Children also suffer from this painful disease. In fact, the incidence of pediatric kidney stones has nearly doubled in the past decade. Annual medical costs of kidney and ureteral stones treatment are estimated at over $10B, with 3 million visits to healthcare professionals per year. Existing treatments have limited effectiveness or require invasive procedures. Ureteral stones in particular account for approximately half of stone cases seen by physicians. Stones over 5 mm have <50% likelihood of passage without intervention. The two most common interventions available for breaking large ureteral stones are Shock Wave Lithotripsy (SWL) and Ureteroscopic Lithotripsy (URS). Both approaches require general anesthesia and SWL often employs the use of ionizing radiation for image guidance. The main benefit of SWL relative to URS is that it is non-invasive. However, it has sub-par efficacy for ureteral stones and the physician often does not know whether the procedure was successful in real time. Only at the 2 or 4 week follow-up visit can the physician assess if the stone were broken successfully. Additionally, the machines are not widely available due to the very high (~ $500K) capital equipment cost. URS is considered more effective than SWL, especially for stones located in the distal ureter. However, URS is an invasive procedure that requires specialized training, general anesthesia, and the use of ureteral stents that are highly uncomfortable to patients. The irritation and pain caused by these stents often leads to additional unscheduled procedures and ER visit and is the main reason why patients prefer not to have URS.
SonoMotion is developing a novel non-invasive stone breaking procedure called Break Wave that has the potential to be a much faster, safer, more effective, and cheaper approach than SWL or URS. Break Wave applies broadly focused bursts of ultrasound, rather than shock waves, to break stones by exciting natural resonances in the stone. Promising initial bench-top and in-vivo animal studies demonstrate the ability to break all common stone types, using ~10 times lower focal pressure amplitude and with the unique ability to control fragment size. Leveraging enhanced ultrasound imaging technologies for stones also being developed by SonoMotion, Break Wave will be entirely ultrasound image guided, thus eliminating radiation exposure from the procedure. The prospect of eliminating ionizing radiation and general anesthesia from lithotripsy procedures in adults and children will be a game changer for Earth and space bound patients. From a societal and health economics perspective, Break Wave is well-aligned with the new quality-based healthcare paradigm. It will reduce healthcare visits and costly repeat stone breaking procedures by breaking the stones into smaller fragments consistently and providing reliable end detection. The elimination of general anesthesia and X-ray C-arm will greatly reduce the cost of the procedure itself and the capital equipment needed.

An additional societal benefit of Break Wave is in how it will help children with ureteral stones. The tools available to treat ureteral stones are often oversized for the pediatric population and may lead to overexposure of ionizing radiation, multiple anesthetics and damaging shock wave pulses. Utereoscopes are simply not designed for children's smaller-diameter ureters. While, SWL is accepted as a first line minimally invasive treatment modality for upper urinary tract stone disease in the pediatric population, it is often overpowered for children leading to unintended damage to nearby tissue and has been associated with stunted growth.

Task Progress:	Astronauts are at increased risk of nephrolithiasis (kidney stones) due to microgravity, dehydration, and altered bone metabolism associated with space flight, as well as medications prescribed to mitigate visual impairment and intracranial pressure (VIIP). A stone can cause debilitating pain as it passes, or worse, become obstructing, leading to the progression of urinary tract infection, sepsis, renal failure, and death. Kidney stone disease is a recognized risk for astronauts and is listed in the ExMC Gap Report 4.13 Limited capability to diagnose and treat a renal stone. Break Wave Lithotripsy (BWL or Break Wave) will fragment large kidney stones more quickly, safely, cheaply, and effectively compared to existing FDA approved methods and with the goal of doing so without the need for general anesthesia or ionizing X ray radiation. Eliminating the need for fluoroscopy, general anesthesia, and a surgical facility will have a dramatic impact on the delivery of lithotripsy treatment. Furthermore, the system is projected to be comparable in cost and size to an ultrasound imaging system, making it feasible for use in clinics, emergency rooms, and in outer space. The overall goal of this grant was to design, build, and test a therapeutic ultrasound Break Wave system with specific emphasis on creating a system that is compact, lightweight, and low power. Specifically, the aims of the project were to: • Aim 1: Fabricate an electronic generator and probe for Break Wave treatment with specific focus on a Space oriented system that is compact, lightweight, and power-efficient. • Aim 2: Demonstrate the ability to break kidney stones using the compact generator and probe developed in Aim 1. • Aim 3: Extend generator capabilities to drive both Break Wave and stone pushing based on findings and experience from Aims 1 and 2. The project successfully achieved its goals. Below is a summary of our key achievements: • Successfully redesigned the Break Wave Generator to a size and mass that is similar to computer DVD drive. The Generator volume and mass was reduced by a factor of 10x from its original form. • Reduced the power requirement of the Generator such that it can be driven by a standard 100W medical grade power supply. This power supply is commercially available and is the size of a pack of cards. • Extended the Generator pulse generation capabilities to both reposition and break stones. • Simulated, designed, and fabricated a four channel Break Wave therapeutic probe that is 85 cm in diameter and which can accommodate a standard phased array imaging probe for procedure guidance. • Redesigned the Break Wave architecture to allow for integration with the NASA FUS or any commercially available diagnostic ultrasound system. Meaning, we should be able to integrate with any diagnostic ultrasound system that NASA chooses to fly on long duration space mission.
Bibliography Type:	Description: (Last Updated:)